

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA0301	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	12024	Date:	24.07.2026

Code of Conduct

I Sharatha sree R(192421033) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sharatha sree R

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

100%

Name: Sharath Sree R

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Sub Code: ITA0201

Sub Name: Mobile
Computing

1.

$$\text{total spectrum} = 45 \text{ MHz} = 45,000 \text{ KHz}$$

$$\text{full duplex channel needs} = 2 \times 25 = 50 \text{ KHz}$$

$$\text{Channels} = \frac{45,000}{50} = \boxed{900}$$

2.

$$\text{Channels per cell} = \frac{\text{total Channel}}{N}$$

$$= \frac{560}{7} = \boxed{80}$$

3.

$$\text{total Voice channels} = \frac{\text{Carrier} \times \text{slot}}{\text{Carrier}}$$

$$= 100 \times 8 = \boxed{800}$$

4.

$$\text{Power ratio} = \frac{80}{4} = 20$$

$$\text{Radio in dB} = 10 \times \log(10(20))$$

$$= 13.01 \text{ dB}$$

$$= \boxed{13 \text{ dB}}$$

5. $C/I \text{ (dB)} = \text{signal dB} - \text{interference (dBm)}$
 $= -60 - (-92) = -60 + 92$
 $= \boxed{32}$

6. $\text{Total channels} = \frac{42,000 \text{ KHz}}{300 \text{ KHz}} = \boxed{140}$

$\text{Channels per cell} = \frac{140}{7} = \boxed{20}$

7. Received power $\propto 1/d^n$. doubling distance with n
 $n = 4 \times \text{reduces power by } 2^4$
 $= 16 \times$

$\text{New power} = \frac{64}{16} = \boxed{4 \text{ W}}$

8. With $n=2$ (inverse square law) doubling radius
 requires power increased by $2^2 = 4 \times$ to maintain
 same edge SN.

$\therefore \boxed{4 \times}$

9. $\text{total channels in cluster} = \text{cluster size} \times \text{channels cell}$
 $= 4 \times 15 = \boxed{60}$

10. $V = 100 \text{ km/h} = 27.8 \text{ m/s}$

$\text{Doppler shift} = V \times \frac{f}{c} = \frac{27.8 \times 900 \times \frac{10}{6}}{3 \times 10^8}$

$$83.3 \text{ Hz}$$

$$\text{slot duration} = \frac{\text{frame time}}{\text{slots}} = \frac{5 \text{ ms}}{8}$$

$$= \boxed{0.625 \text{ ms}}$$

$$12. \text{ Carriers} = \frac{\text{total spectrum}}{\text{spacing}}$$

$$= \frac{24,000 \text{ KHz}}{200 \text{ KHz}}$$

$$= \boxed{120}$$

$$13. \text{ Tx power in dBm} = 10 \log_{10} (8000 \text{ mW})$$

$$= 39.03 \text{ dBm}$$

$$\text{Received power} = 39.03 - 33 = 6.03$$

$$\boxed{\text{dBm} \approx 6 \text{ dB}}$$

14. Using the log-distance path loss model referenced to 1m (consistent with $= 120 \text{ dB}$) at 2 km for a nearly frequency band scaling the distance to 2.5 km given approximate free-space path loss of 123 dB.

$$\therefore \boxed{123 \text{ dB}}$$

$$15. \text{ total capacity} = \frac{\text{cells}}{\text{Cells} \times \text{cells}} = 250 \times 6$$

$$= \boxed{1500}$$

$$\begin{aligned}
 17. \text{ total bandwidth} &= \frac{\text{channel} \times \text{bandwidth}}{\text{channel}} \\
 &= 150 \times 25 \text{ KHz} = 3750 \text{ KHz} \\
 &= \boxed{3.75 \text{ MHz}}
 \end{aligned}$$

$$18. \left. \begin{array}{l} \text{Number of guard bands between} \\ 30 \text{ bands} \end{array} \right\} = 29$$

$$\begin{aligned}
 \text{Wasted bandwidth} &= 29 \times 40 \text{ KHz} \\
 &= 1160 \text{ KHz} \\
 &= \boxed{1.16 \text{ MHz}}
 \end{aligned}$$

$$19. \text{ Speed} = 90 \text{ km/h} = 25 \text{ m/s}$$

$$\begin{aligned}
 \text{Cell diameter} &= 2 \times 1.5 \text{ km} \\
 &= \boxed{3000 \text{ m}}
 \end{aligned}$$

$$\text{time per cell} = \frac{3000}{25} = \boxed{120 \text{ s}}$$

$$\text{Handoffs / hour} = \frac{3600}{120} = \boxed{30}$$

$$\begin{aligned}
 20. \text{ total users} &= \text{regional centers} \times \text{base station / center} \\
 &\quad \times \text{users / base station} \\
 &= 8 \times 40 \times 250 \\
 &= 80,000
 \end{aligned}$$

$$\begin{aligned}
 21. \text{ Approximate cell area} &= R^2 \\
 &= (0.4 \text{ km})^2 \\
 &= \boxed{0.16 \text{ km}^2}
 \end{aligned}$$

$$\text{nbr of cells} = \frac{400}{0.16} = \boxed{2500}$$

$$\text{channels} = \frac{\text{total spectrum}}{\text{channel width}} = \frac{6000 \text{ KHz}}{30 \text{ KHz}} = \boxed{200}$$

$$23. \text{ total call minutes} = 1200 \times 4 \times 2.5 = 12000 \text{ min/hr}$$

$$\text{traffic in erlangs} = \frac{1200}{60} = \boxed{200}$$

$$24. U_{ms} = \frac{\text{total bandwidth}}{\text{bandwidth per user}} = \frac{40000 \text{ KHz}}{400 \text{ KHz}} = \boxed{100}$$

$$25. \text{ total capacity} = \frac{U_{ms}}{\text{cell} \times \text{cells}} = 600 \times 1200 = \boxed{720,000}$$

$$26. \text{ Paging operation} = \frac{\text{total cells}}{\text{cells operations}} = \frac{200}{4} = \boxed{50}$$

$$27. \text{ Offered traffic} = \text{arrival rate} \times \text{avg duration} = 40000 \times 3 = 1,20,000 \text{ erlangs.}$$

$$\text{Excess capacity} = 120000 - 100000 = 20000$$

$$\text{fraction blocked} = \frac{20000}{120000} = \boxed{0.17}$$

$$\text{Roaming Users} = 15\% \times 600,000$$

$$= \boxed{90,000}$$

$$\text{Utilization} = \frac{\text{peak traffic}}{\text{Processing rate}} = \frac{9000}{12000}$$

$$= 0.75 = \boxed{75\%}$$

$$\text{total channels} = \frac{\text{bandwidth}}{\text{Channelwidth}}$$

$$= \frac{48000}{200} = 240$$

$$\text{Channels per cell} = \frac{240}{4} = \boxed{60}$$

$$\text{Channels} = \frac{\text{total spectrum}}{\text{Channel width}} = \frac{30000}{200}$$

$$= \boxed{150}$$

$$\text{Successful handoffs} = 99\% \times 2500$$

$$= \boxed{2475}$$

$$\text{Tx power in dBm} = 10 \log_{10} (12000 \text{ mW})$$

$$\approx 40.79 \text{ dBm}$$

$$\text{received power} = 40.79 - 45$$

$$\approx -4.21 \text{ dBm}$$

$$\approx \boxed{-4 \text{ dBm}}$$

$$\text{Total capacity} = \frac{U_{\text{sys}}}{\text{Cell} \times \text{cells}} = 900 \times 7 = \boxed{6300}$$

$$35. \text{ Slot duration} = \frac{\text{frame time}}{\text{slots}} = \frac{4.8 \text{ ms}}{8}$$

$$= \boxed{0.6 \text{ ms}}$$

$$36. v = 150 \text{ km/h} = 41.67 \text{ m/s}$$

$$\text{Doppler shift} = v \times \frac{f}{c} = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8}$$

$$= \boxed{250 \text{ Hz}}$$

$$37. \text{ Offered traffic} = 70,000 \times 4 = 280,000 \text{ Erlangs.}$$

$$\text{Excess} = 280,000 - 250,000 = 30,000$$

$$\text{fraction blocked} = \frac{30,000}{280,000}$$

$$\approx \boxed{0.11}$$

$$38. \text{ total using cluster} = \frac{U_{\text{sys}}}{\text{cell} \times \text{cells in cluster}}$$

$$= 600 \times 9 = \boxed{5400}$$

$$39. \text{ Calls dropped / sec} = 0.4\% \times 601000$$

$$= 240$$

$$40. \text{ Path loss (dB)} = \text{TX power (dBm)} - \text{RX power (dBm)}$$

$$= 43 - (-87)$$

$$= \boxed{130 \text{ dB}}$$

$$41. \text{ Total coverage} = \frac{\text{coverage}}{\text{Station} \times \text{no of station}}$$

$$= 2.5 \times 150 = \boxed{375 \text{ km}^2}$$

$$42. \text{ Total channels} = \frac{481000}{200}$$

$$= 240$$

$$\text{Channels per cell} = \frac{240}{6} = \boxed{40}$$

$$43. \text{ Paging cycles} = \frac{\text{total users}}{\text{Users per cycle}} = \frac{1501000}{6000} = \boxed{25}$$

$$44. \text{ Handoffs per call} = \frac{\text{call duration}}{\text{handoff interval}} = \frac{9}{3} = \boxed{3}$$

$$45. \text{ SNR (dB)} = \text{signal (dBm)} - \text{noise (dBm)}$$

$$= -85 - (-115)$$

$$= -85 + 115$$

$$= \boxed{30 \text{ dB}}$$

$$\begin{aligned}
 \text{Minimum bandwidth} &= U_{ms} \times \text{rate} / \text{user} \\
 &= 250 \times 150 \text{ kbps} \\
 &= 37,500 \text{ kbps} \\
 &= \boxed{37.5 \text{ Mbps}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total call} &= \frac{U_{ms} \times \text{calls}}{\text{hour} \times \text{avg duration}} = 150 \times 4 \times 2.5 \\
 &= 15,000 \text{ min}
 \end{aligned}$$

$$\text{Traffic in Erlangs} = \frac{15,000}{60} = \boxed{250}$$

$$\begin{aligned}
 \text{48. Total bandwidth} &= \text{Channel} \times \text{bandwidth} / \text{Channel} \\
 &= 250 \times 12 \text{ KHz} \\
 &= 3000 \text{ KHz} = \boxed{3 \text{ MHz.}}
 \end{aligned}$$

$$\begin{aligned}
 \text{49. Total throughput} &= \text{Connection} \times \text{rate} / \text{Connection} \\
 &= 1500 \times 600 \text{ kbps} \\
 &= 900,000 \text{ kbps} = \boxed{900 \text{ Mbps}}
 \end{aligned}$$

$$\text{50. Area per tower} = \pi R^2 = \pi \times 4^2 \approx 50.27 \text{ km}^2$$

$$\text{towers needed} = \frac{350}{50.27} \approx 6.96 \approx \boxed{7}$$

$$\begin{aligned}
 \text{16. Active calls supported} &= \text{arrival rate} \times \text{avg duration} \\
 &= 4000 \text{ calls / sec} \times 180 \\
 &= \boxed{720,000}
 \end{aligned}$$